

Daiki Koike

koike.d@northeastern.edu | +1 415 740 8123 | Oakland, CA | [linkedin.com/in/daikikoike](https://www.linkedin.com/in/daikikoike) | github.com/Daiki007121

Open to: Summer 2026 Internship | Full-time from December 2026

EDUCATION

Northeastern University – Khoury College of Computer Sciences, Oakland, CA Sep 2023 – Present

Align MS in Computer Science Candidate, GPA: 3.85 / 4.0 Expected Graduation: December 2026

Relevant Coursework: AI-Assisted Software Engineering, Foundations of AI, Algorithms, Software Engineering, Object-Oriented Design, Data Structures & Algorithms, Web Development, Discrete Structures

San Francisco State University, San Francisco, CA Aug 2018 – May 2022

B.A. Physics, Magna Cum Laude, GPA: 3.83 / 4.0

TECHNICAL KNOWLEDGE

Languages: Python, JavaScript, TypeScript, HTML, CSS

AI/LLM Tooling: Claude Code (custom skills, MCP servers, sub-agents, hooks, CLAUDE.md), Anthropic API, Gemini Live API

Frameworks & Tools: Next.js, React, Node.js, Express, Supabase, Vercel, MongoDB, Git

Methodologies: TDD, Agile, CI/CD with GitHub Actions

PROJECTS

MaskLM — Privacy-First PII-Masking Middleware for LLMs Mar 2026 – Apr 2026

- Built privacy-first PII-masking middleware for LLMs with Next.js, Supabase, and FastAPI; led web-app architecture decision over PyPI to support non-technical end users.
- Designed stateless backend with zero PII persistence; eliminated server-side attack surface by removing all database, session, and request-body logging.
- Shipped /tdd-feature custom Claude Code skill and GitHub MCP integration to automate the RED-GREEN-REFACTOR cycle; deployed to production with Microsoft Presidio for PII detection.

AI-Powered Mom Test Interviewer — Real-Time Voice AI Feb 2026 – Mar 2026

- Built real-time voice AI application using Next.js 14, Supabase, and Gemini Live API to automate unbiased Mom Test-style customer interviews.
- Designed a three-tier system (Next.js + Node.js WebSocket relay + Supabase) and solved cross-device audio pipeline bugs (sample rate mismatch, RMS-based turn detection).
- Achieved **97%** test coverage (**243** tests) with TDD; configured GitHub Actions CI/CD and structured-JSON insight pipeline for validation reports.

3.6 Million Routes, One Perfect Day — A TSP Study on Real Bay Area Transit Data Apr 2026

- Solved a Traveling Salesman Problem over 11 Bay Area landmarks (Northeastern Oakland campus + 10 SF/East Bay spots) using real Google Maps transit times.
- Compared four algorithms: BFS exhaustive enumeration ($10! = 3.6\text{M}$ permutations, optimal baseline), Greedy nearest-neighbor heuristic, Random (1000 permutations as upper-bound sanity check), and A* with an MST-based admissible heuristic.
- Achieved optimal solution with A* using $\sim 2000\times$ less work than BFS; implemented and benchmarked all four algorithms in Python.

EXPERIENCE

Japan Blockchain Association, Tokyo, Japan Nov 2022 – Jun 2023

Intern

- Conducted background investigations on companies and individuals applying to join the association, verifying legitimacy and screening for compliance risks and organized-crime affiliations.
- Organized and participated in Web3 hackathons as part of a 4-member team, facilitating collaboration between developers and industry partners.

CSME – Center for Science and Math Education, SFSU, San Francisco, CA Nov 2021 – May 2022

Teacher Fellow

- Tutored high school students in mathematics and physics through a competitive fellowship program, providing one-on-one academic support for **50+** hours.
- Participated in teacher training workshops focused on effective instructional techniques and student engagement strategies.

HONORS AND AWARDS

- 2023–2026: Regional Campus Scholarship, Northeastern University
- 2022: Phi Beta Kappa Honor Society
- 2022: B.A. Physics, Magna Cum Laude, GPA 3.83 / 4.0, San Francisco State University
- 2021, 2022: Teacher Fellowship, CSME, San Francisco State University
- 2018–2022: Dean's List, eight consecutive terms, San Francisco State University